Cheng CHEN

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EDUCATION

Imperial College London (IC) | London, UK — 09/2025- Present (Expected Graduation: 09/2026)
M.S., Applied Computational Science and Engineering

Donghua University (DHU) ("211 Project" university) | Shanghai, China — 09/2021-06/2025
B.S., Data Science and Big Data Technology **GPA:** 90.1/100

INTERNSHIP

1. Artificial Intelligence Quantitative Research Lab | Shanghai, China | Research Assistant — 02/2025-08/2025

- Developed a Mixture-of-Experts–inspired **ensemble framework** that enhanced localized signal representation across diverse alpha clusters, effectively addressing low-SNR and high-dimensionality issues in complex factor libraries
- Architected a **Cross-Source Attention Mechanism** utilizing Q-K-V interactions to fuse heterogeneous market data streams (news, social media, search behavior) for downstream prediction tasks
- Fine-tuned **NLP models** for domain-specific financial semantic quantification, enabling sector classification, topic modeling, and real-time market analysis on large-scale, multi-source datasets
- Spearheaded the development of an **Agent System**, enabling both autonomous analytical workflows and on-demand task execution, integrated with modular data pipelines and visualization suite
- Utilized Git for version control and GitHub Actions **(CI/CD)** to automate testing and build processes, ensuring code reusability and production stability

2. Institute of Computing Technology, Chinese Academy of Science | Beijing, China | AI Intern — 08/2024-01/2025

- **Fine-tuned Visual-Language Models** (VLM) for region-level dialogue in remote sensing, enabling **spatial conversation** by textualizing object detection (bounding box) and class data
- Engineered a data pipeline to **enrich text attributes**, including Color (via K-Means clustering), Relative Size (via percentile-based benchmarking), and Relative Location (via grid-mapping)
- Leveraged **Contrastive Learning** (via CLIP) to align cross-domain visual-text representations
- Designed a novel **fusion module** to process multi-dimensional, ultra-heterogeneous inputs, enabling robust Super-Resolution, image alignment, and denoising for multi-spectral images
- Engineered an End-to-End Computer Vision System integrating Edge Enhancement, Super-Resolution Generative Adversarial Networks (SR-GANs), and **downstream tasks** (e.g., Object Detection, Segmentation)

RESEARCH & SELECTED PROJECTS

Authored **academic manuscripts** (currently **under review**). Related project details and code are accessible via my linked personal website/GitHub (see header).

1. Education Big Data Analysis Platform with Integrated Specialized LLM | Group Leader — 02/2024-05/2024

- **National Second Prize**, Chinese Collegiate Computing Competition (4C)
- Engineered LLM assistants by integrating a **Text2SQL** model for structured querying and a **RAG** system for content-based knowledge retrieval

2. Dual-Stream Visual Enhanced 3D Image Segmentation | Research Leader — 08/2023-12/2023

- **1st Place**, Shanghai AI Lab DIMTAIC 2023 Segmentation Competition
- Invited Participant at the 2023 **Health China Sinan Summit**
- Proposed a visual enhancement perception mechanism that collaboratively models global pixel dependencies and multi-scale local texture information, introducing a dual-branch co-attention mechanism for adaptive information scheduling

3. Automatic Annotation System for Cervical Liquid-based Pathology Images | Group Leader — 03/2023-12/2023

- **Awarded: National-Level** project in the Undergraduate Innovation and Entrepreneurship Program
- Led development on two primary components: a model for precise cervical cell detection and a specialized medical large language model (LLM)

4. Multi-modal Solution: Deepfake Detection and the Source Identification | Co-first Author — 07/2023-10/2023

- A Multi-Modal Deepfake Detection Framework, integrating a Cross-Modal Attention
- Relevant Publication: Zheng, Y., **Chen, C.***, Zhou, Xu., & Hu, J. (Co-first Author). Multi-Modal Solution: Deepfake Detection and the Source Identification. PMBDA 2023